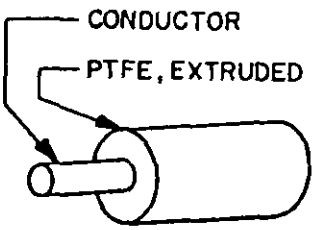
MIL-W-16878/25(NAVY)
10 August 1981
SUPERSEDING
MIL-W-16878/4A(NAVY) (IN PART)
5 July 1961
(See supersession data herein)

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE (PTFE) INSULATED, 260°C, 600 VOLTS, EXTRUDED INSULATION

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.



SH 11875

FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/ 3/	Coating		Diameter (inch)	
					Min	Max
32	1 X 32	Copper	Nickel	0.089	0.025	0.033
32	1 X 32	H.S.C.A.	Nickel	.089	.025	.033
32	1 X 32	C.C. Steel	Nickel	.089	.025	.033

See footnotes at end of table.

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:31Z
Check the source to verify that this is the current version before use.

TABLE I. Construction details. - Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/ 3/	Coating		Diameter (inch)	
					Min	Max
32	7 X 40	Copper	Nickel	0.010	0.026	0.034
32	7 X 40	H.S.C.A.	Nickel	.010	.026	.034
30	1 X 30	Copper	Nickel	.0100	.026	.034
30	1 X 30	H.S.C.A.	Nickel	.0100	.026	.034
30	1 X 30	C.C. Steel	Nickel	.0100	.026	.034
30	7 X 38	Copper	Nickel	.012	.028	.036
30	7 X 38	H.S.C.A.	Nickel	.012	.028	.036
28	1 X 28	Copper	Nickel	.0126	.029	.037
28	1 X 28	H.S.C.A.	Nickel	.0126	.029	.037
28	1 X 28	C.C. Steel	Nickel	.0126	.029	.037
28	7 X 36	Copper	Nickel	.015	.031	.039
28	7 X 36	H.S.C.A.	Nickel	.015	.031	.039
26	1 X 26	Copper	Nickel	.0159	.032	.040
26	1 X 26	H.S.C.A.	Nickel	.0159	.032	.040
26	1 X 26	C.C. Steel	Nickel	.0159	.032	.040
26	7 X 34	Copper	Nickel	.019	.035	.043
26	7 X 34	H.S.C.A.	Nickel	.019	.035	.043
26	19 X 38	Copper	Nickel	.020	.035	.043
26	19 X 38	H.S.C.A.	Nickel	.020	.035	.043
24	1 X 24	Copper	Nickel	.0201	.036	.044
24	1 X 24	H.S.C.A.	Nickel	.0201	.036	.044
1/ 24	1 X 24	C.C. Steel	Nickel	.0201	.036	.044
24	7 X 32	Copper	Nickel	.024	.040	.048
24	7 X 32	H.S.C.A.	Nickel	.024	.040	.048
24	19 X 36	Copper	Nickel	.025	.040	.048
24	19 X 36	H.S.C.A	Nickel	.025	.040	.048
22	1 X 22	Copper	Nickel	.0254	.041	.049
22	1 X 22	H.S.C.A.	Nickel	.0254	.041	.049
1/ 22	1 X 22	C.C. Steel	Nickel	.0254	.041	.049
22	7 X 30	Copper	Nickel	.030	.046	.054
22	7 X 30	H.S.C.A.	Nickel	.030	.046	.054
22	19 X 34	Copper	Nickel	.032	.046	.054
22	19 X 34	H.S.C.A.	Nickel	.032	.046	.054
20	1 X 20	Copper	Nickel	.0320	.048	.056
20	1 X 20	H.S.C.A.	Nickel	.0320	.048	.056
20	7 X 28	Copper	Nickel	.038	.054	.062
20	7 X 28	H.S.C.A.	Nickel	.038	.054	.062
20	19 X 32	Copper	Nickel	.040	.054	.062
20	19 X 32	H.S.C.A.	Nickel	.040	.054	.062

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:31Z
Check the source to verify that this is the current version before use.

TABLE I. Construction details. - Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/ 3/	Coating		Diameter (inch)	
					Min	Max
18	1 X 18	Copper	Nickel	0.0403	0.056	0.066
18	7 X 26	Copper	Nickel	.048	.064	.074
18	19 X 30	Copper	Nickel	.050	.064	.074
16	1 X 16	Copper	Nickel	.0508	.067	.081
4/ 16	19 X 29	Copper	Nickel	.057	.073	.087
14	19 X 27	Copper	Nickel	.072	.086	.100
12	19 X 25	Copper	Nickel	.091	.107	.121
12	37 X 28	Copper	Nickel	.089	.105	.119
10	37 X 26	Copper	Nickel	.111	.127	.141

1/ Inactive for new design.
2/ H.S.C.A. stands for high strength copper alloy.
3/ C.C. Steel stands for copper clad steel.
4/ Under MPD-1506, the 16 AWG size was heavier.

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 3.4 kilovolts (kV).
Impulse dielectric test: 6.5 kV.
Tank test: 2.0 kV.
Dielectric withstanding voltage: 2.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C
K = 50,000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 65°C ± 1°C (see table II).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:31Z
Check the source to verify that this is the current version before use.

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
32 through 16	1
14 through 10	2

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Applicable.
Heat resistance: Condition at 290°C.
Flammability: Not applicable.
Soldering: Applicable.
Heat aging: Not applicable.
Insulation tensile strength: 3000 pounds force per square inch (minimum).
Insulation elongation: 150 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

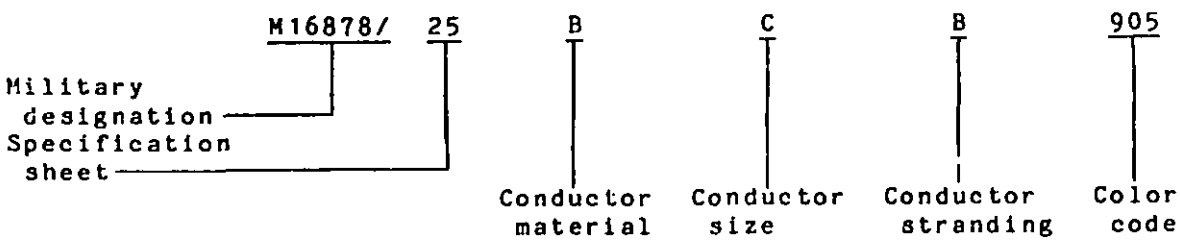
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:31Z
Check the source to verify that this is the current version before use.

TABLE III. Put-up tabulation.[1/]

AWG size	Percentage of order	Length (ft) one continuous length	Maximum no. of lengths (per spool or reel)	
			500 ft	1000 ft
32 to 22	25 25 50	Under 75 75 - 300 301 or more	3	4
20 to 16	Unacceptable 20 50 30	Under 75 75 - 100 101 - 300 301 or more	4	5
14 to 10	Unacceptable 40 50 10	Under 50 50 - 100 101 - 300 301 or more	5	6

[1/] Each spool or reel shall be marked accordingly with the length in feet and location of each piece.

Military part number (see MIL-W-16878):



Supersession data: The wire covered by this specification sheet supersedes MIL-W-16878/4A(NAVY), type E, 260°C rated, with extruded insulation.

Review activities:
Navy - AS, EC
User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N300-24)

☆U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/5797

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:31Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. *If block 7 is filled out, an* acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Sea System Command (SEA 3112)
Department of the Navy
Washington, DC 20362

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil as of 2026-10-18T09:31Z
Check the source to verify that this is the current version before use.